

Problem

Rework Example 17.14 if the low-pass filter is replaced by the ideal bandpass filter shown in Fig. 17.46.

Figure 17.46:



Example 17.14:

If the sawtooth waveform in Fig. 17.45(a) is applied to an ideal low-pass filter with the transfer function shown in Fig. 17.45(b), determine the output.

Figure 17.45(a):

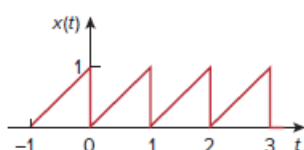
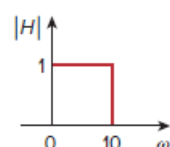


Figure 17.45(b):



Solution:

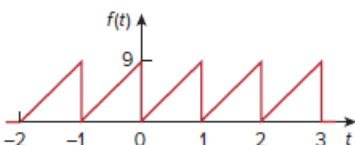
The input signal in Fig. 17.45(a) is the same as the signal in Fig. 17.9. From Practice Prob. 17.2, we know that the Fourier series expansion is

$$x(t) = \frac{1}{2} - \frac{1}{\pi} \sin \omega_0 t - \frac{1}{2\pi} \sin 2\omega_0 t - \frac{1}{3\pi} \sin 3\omega_0 t - \dots$$

where the period is $T = 1$ s and the fundamental frequency is $\omega_0 = 2\pi$ rad/s. Inasmuch as the corner frequency of the filter is $\omega_c = 10$ rad/s, only the dc component and harmonics with $n\omega_0 < 10$ will be passed. For $n = 2$, $n\omega_0 = 4\pi = 12.566$ rad/s, which is higher than 10 rad/s, meaning that second and higher harmonics will be rejected. Thus, only the dc and fundamental components will be passed. Hence, the output of the filter is

$$y(t) = \frac{1}{2} - \frac{1}{\pi} \sin 2\pi t$$

Figure 17.9:



Step-by-step solution

Refer to Figure 17.45 in the textbook.

The Fourier series expansion for the waveform to Figure 17.45(a).

$$x(t) = \frac{1}{2} - \frac{1}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \sin n\omega_0 t$$

Where the time period is $T = 1 \text{ s}$ and the fundamental frequency is ω_0 .

$$\begin{aligned}\omega_0 &= \frac{2\pi}{T} \\ &= \frac{2\pi}{1} \\ &= 2\pi \text{ rad/s}\end{aligned}$$

The band pass filter passes all the harmonics of the input signal within a band of frequencies $\omega_1 < \omega < \omega_2$ centered around ω_c .

Where,

$$\omega_1 = 15 \text{ rad/sec}$$

$$\omega_2 = 35 \text{ rad/sec}$$

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For $n = 1$

$$\begin{aligned}n\omega_0 &= 1(2\pi) \\ &= 6.2831 \text{ rad/s}\end{aligned}$$

This component is not within the band of frequencies $15 < \omega < 35$. So, this harmonic will be rejected.

For $n = 2$

$$\begin{aligned}n\omega_0 &= (2)(2\pi) \\ &= 12.5663 \text{ rad/s}\end{aligned}$$

This component is not within the band of frequencies $15 < \omega < 35$. So, this harmonic will be rejected.

For $n = 3$

$$\begin{aligned}n\omega_0 &= (3)(2\pi) \\ &= 18.84955 \text{ rad/s}\end{aligned}$$

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This component is within the band of frequencies $15 < \omega < 35$. So, this harmonic will be allowed.

For $n = 4$

$$\begin{aligned} n\omega_0 &= (4)(2\pi) \\ &= 25.13274 \text{ rad/s} \end{aligned}$$

This component is within the band of frequencies $15 < \omega < 35$. So, this harmonic will be allowed.

For $n = 5$

$$\begin{aligned} n\omega_0 &= (5)(2\pi) \\ &= 31.41592 \text{ rad/s} \end{aligned}$$

This component is within the band of frequencies $15 < \omega < 35$. So, this harmonic will be allowed.

For $n = 6$

$$\begin{aligned} n\omega_0 &= (6)(2\pi) \\ &= 37.39911 \text{ rad/s} \end{aligned}$$

This component is not within the band of frequencies $15 < \omega < 35$. So, this harmonic will be rejected.

All harmonic components having frequency greater than 35 rad/sec are not allowed by this filter.

Step 4 of 4

Therefore, only three components of frequencies $3\omega_0$, $4\omega_0$, $5\omega_0$ are passed through this band pass filter.

Therefore, the output of the filter $y(t)$ is $\boxed{-\frac{1}{3\pi} \sin 3\omega_0 t - \frac{1}{4\pi} \sin 4\omega_0 t - \frac{1}{5\pi} \sin 5\omega_0 t}$.